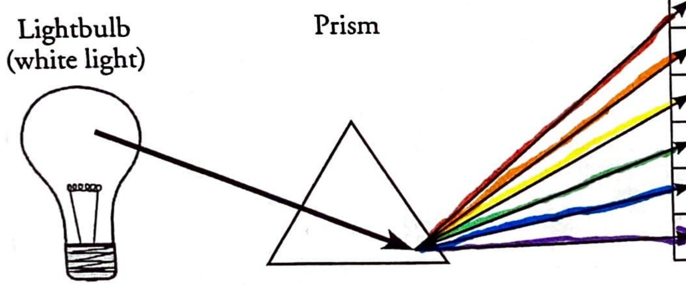
Electron Energy and Light

How does light reveal the behavior of electrons in an atom?

Why?

From fireworks to stars, the color of light is useful in finding out what's in matter. The emission of light by hydrogen and other atoms has played a key role in understanding the electronic structure of atoms. Trace materials, such as evidence from a crime scene, lead in paint or mercury in drinking water, can be identified by heating or burning the materials and examining the color(s) of light given off in the form of bright-line spectra.

Model 1 – White Light



Color	Photon Energy ($\times 10^{-21}$) (J)	Wavelength Range (nm)	Speed (m/s)
Reds	269–318	625–740	3.00×10^8
Oranges	318–337	590–625	3.00×10^8
Yellows	337–352	565–590	3.00×10^8
Greens	352–382	520–565	3.00×10^8
Blues	382–452	440–520	3.00×10^8
Violets	452–523	380–440	3.00×10^8

1. Trace the arrows in Model 1 and shade in the table with colored pencils where appropriate.
2. What happens to white light when it passes through a prism?

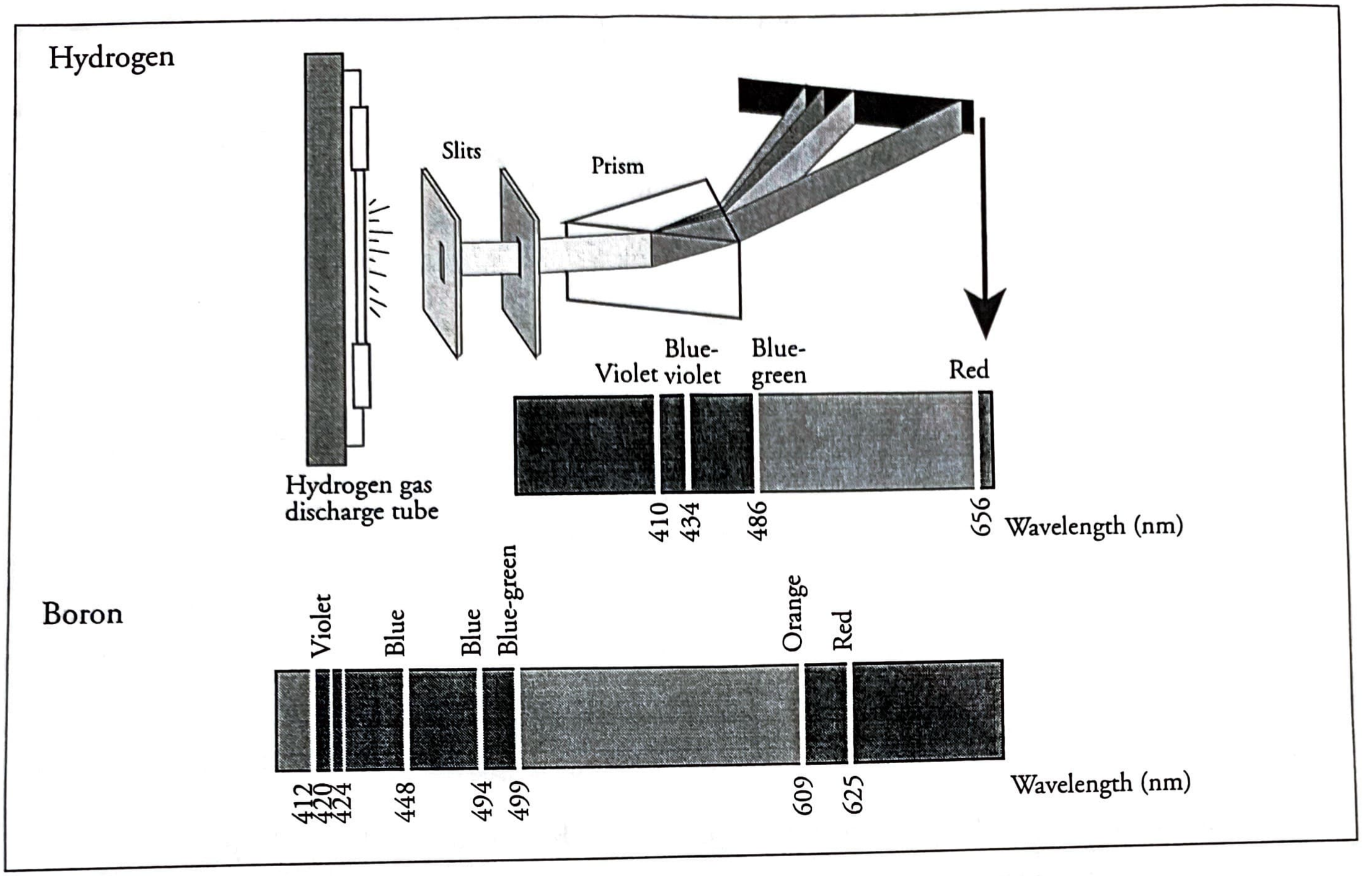
   ‡ When white light passes through a prism it shows all th the co it's ma out of.
3. Why are the color labels in the table in Model 1 plural (*i.e.*, "Reds" rather than "Red")?
4. Do all colors of light travel at the same speed?
5. Do all colors of light have the same energy? If no, which colors have the highest energy and the least energy, respectively?
6. Consider the light illustrated in Model 1.

   a. Which color corresponds to the longest wavelengths?

   b. Which color corresponds to the shortest wavelengths?

   c. Write a sentence that describes the relationship between wavelength and energy of light.

Model 2 – Emission Spectra for Hydrogen and Boron Atoms



7. Use colored pencils to color the hydrogen and boron spectral lines within their respective spectra in Model 2.

8. List the spectral lines for hydrogen gas by color and corresponding wavelength.

9. The spectral lines for boron were produced using the same method as hydrogen. List three of the colors and corresponding wavelengths for boron's spectral lines as its light passes through a prism.

10. Consider the hydrogen spectrum in Model 2.

    a. Which color of light corresponds to the shortest wavelength?

    b. Which color of light corresponds to the longest wavelength?